

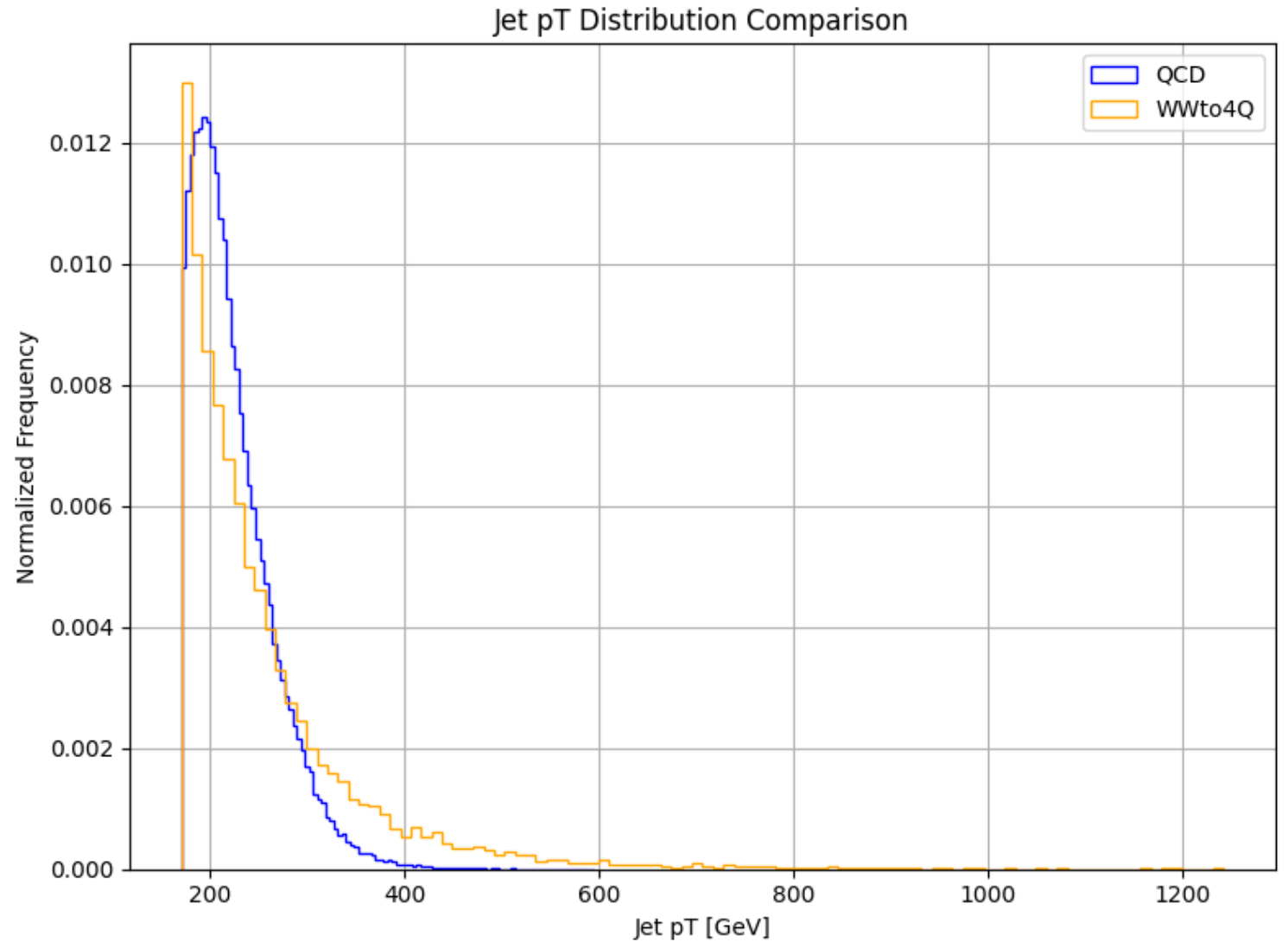
Week 4.5

8-July to 11-July

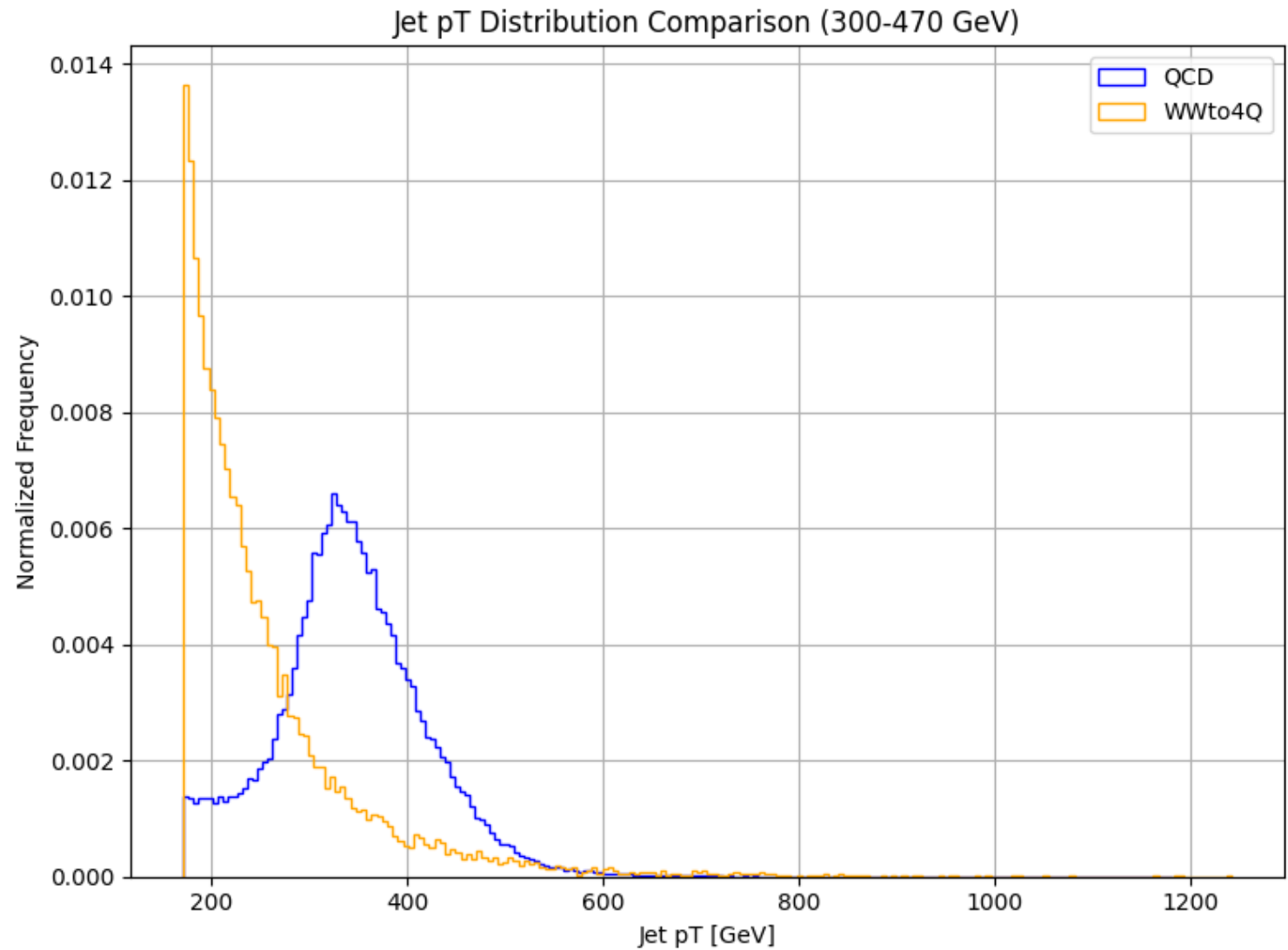
Goals

- Train a couple models on different pT slices
- Background: QCD
- Signal: WWto4Q

pT: 200-
300 GeV

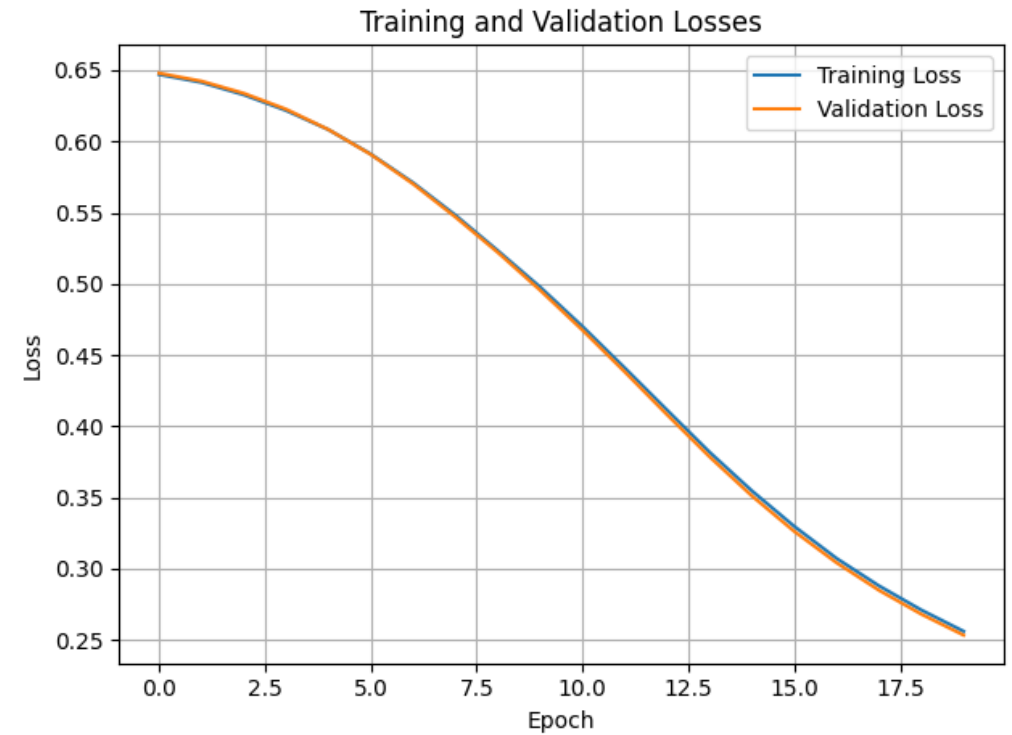
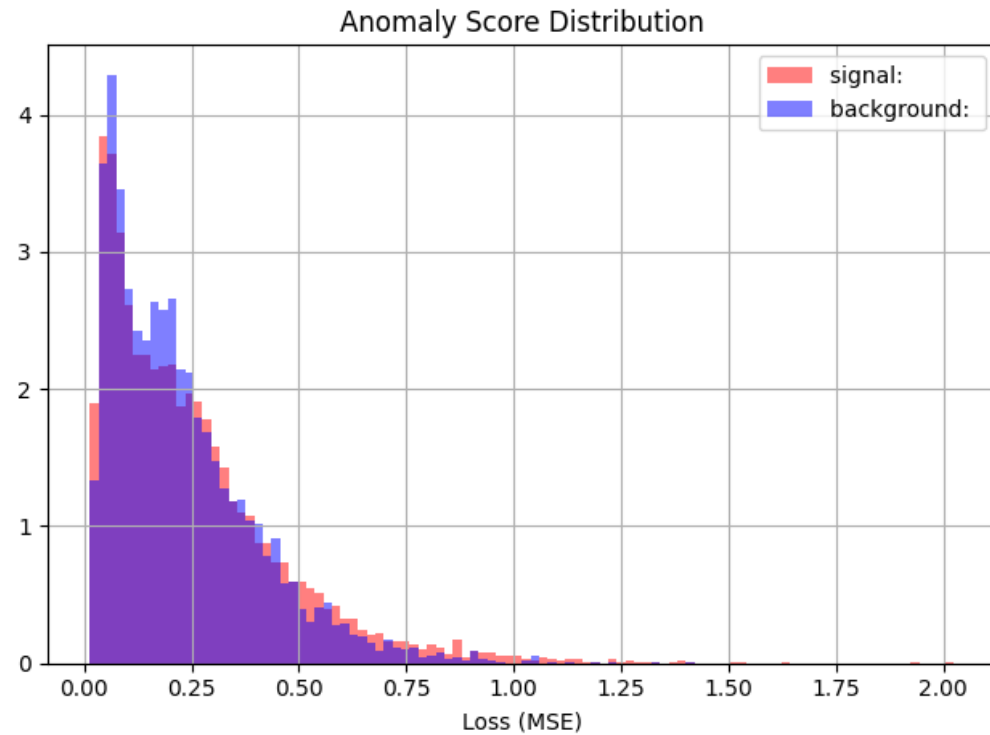


pT: 300-
400 GeV



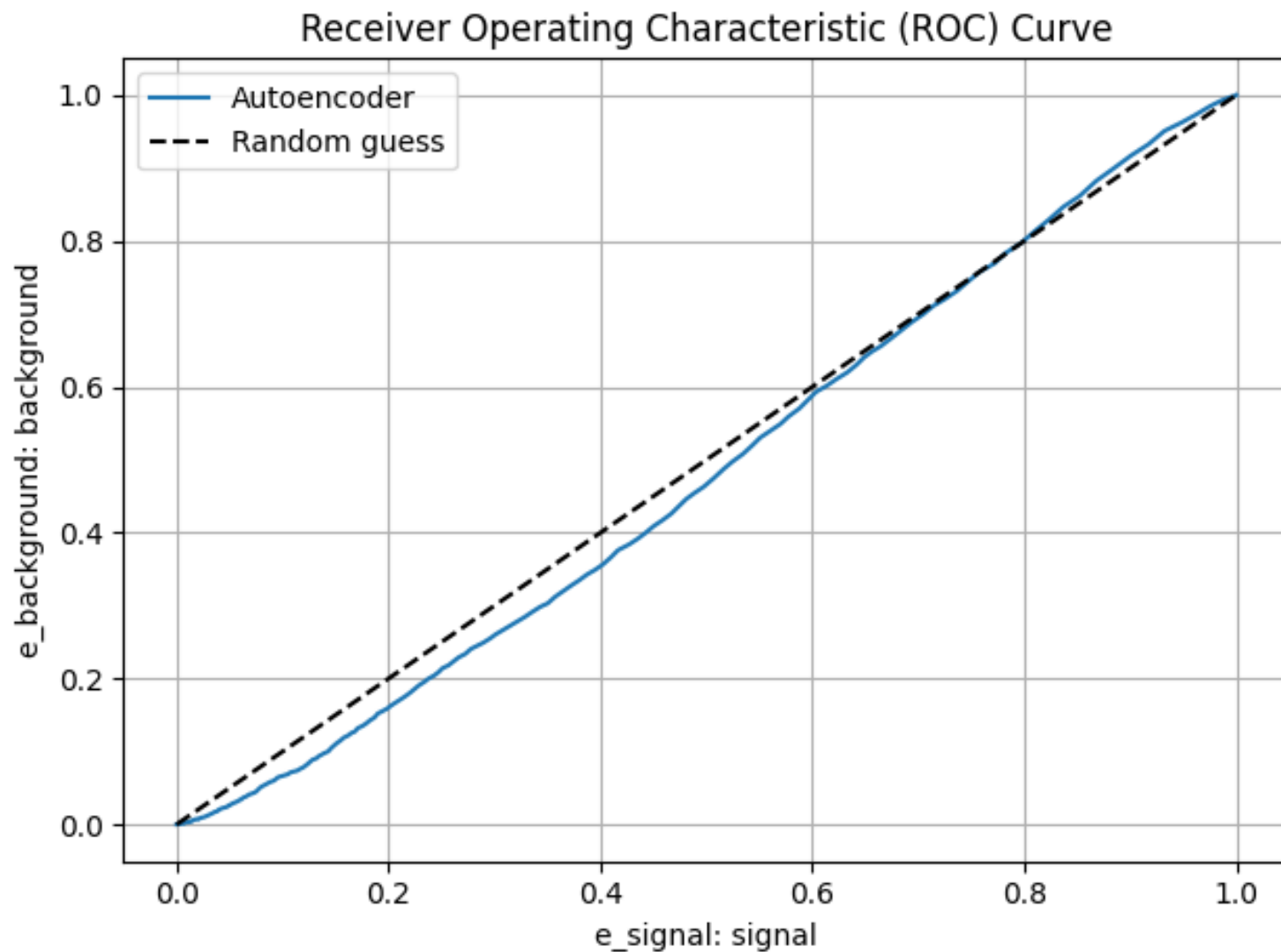
pT: 200-300 GeV

- Almost no separation



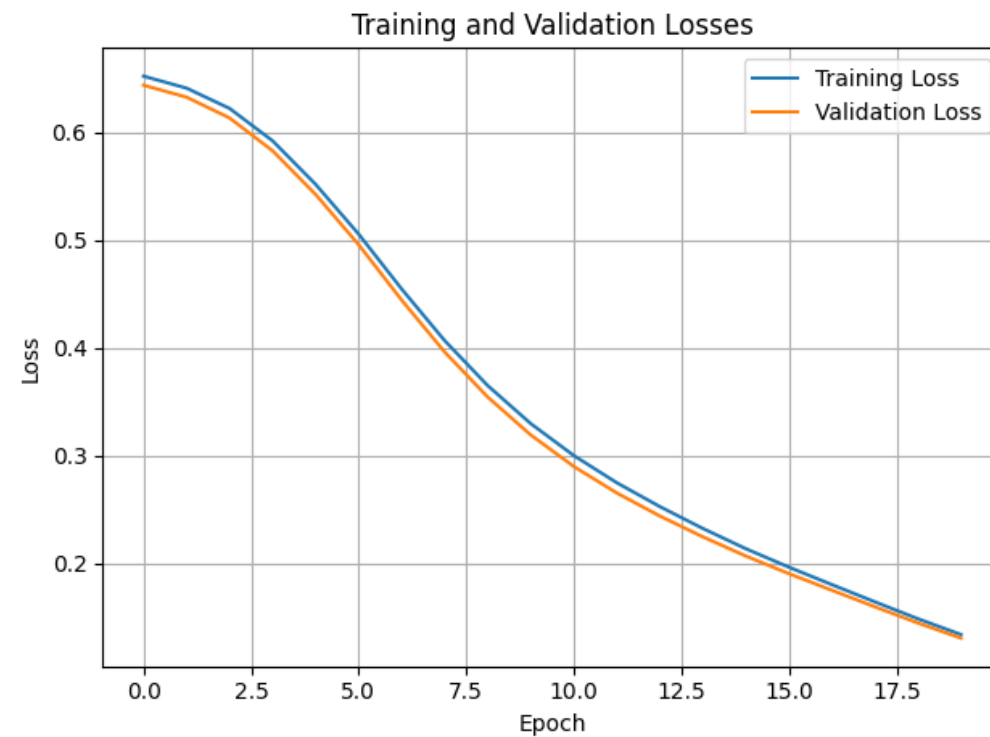
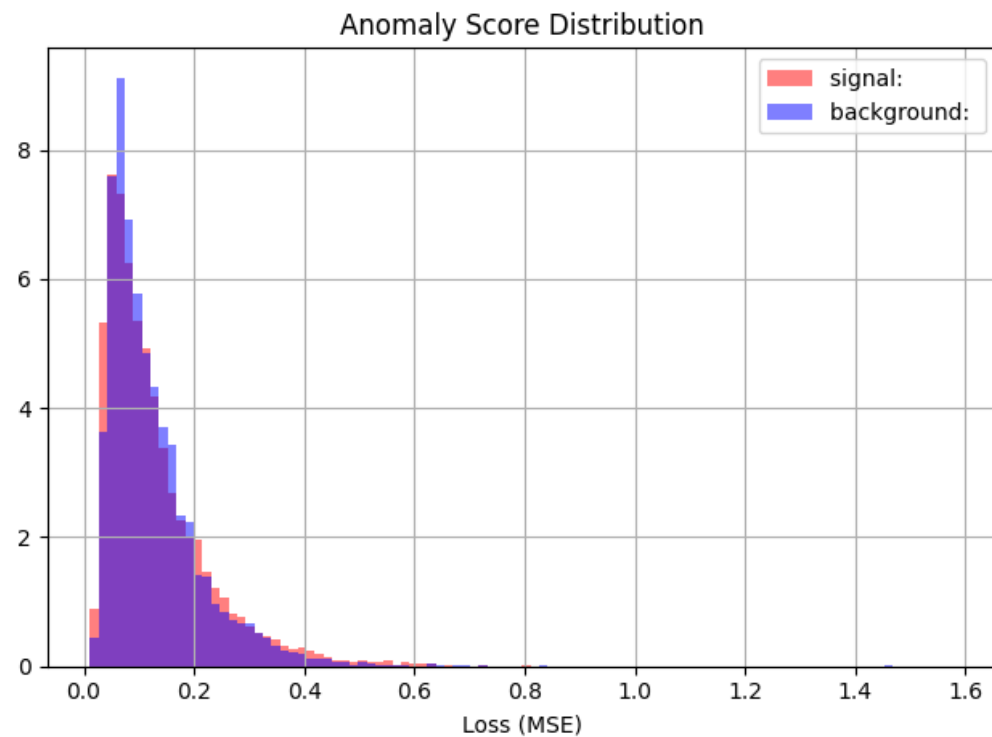
ROC

AUC = 0.520



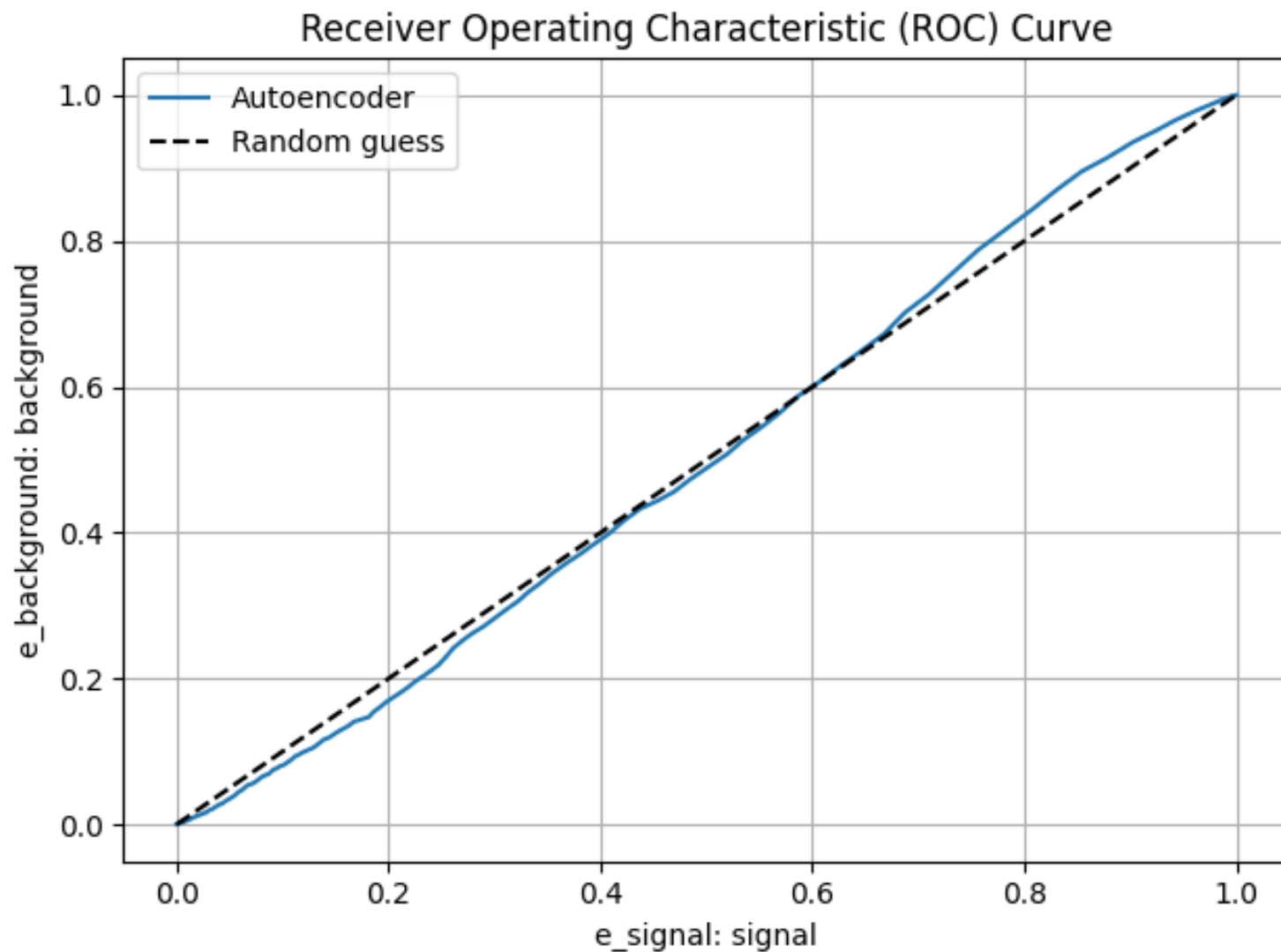
pT: 300-400 GeV

- Again, almost no separation



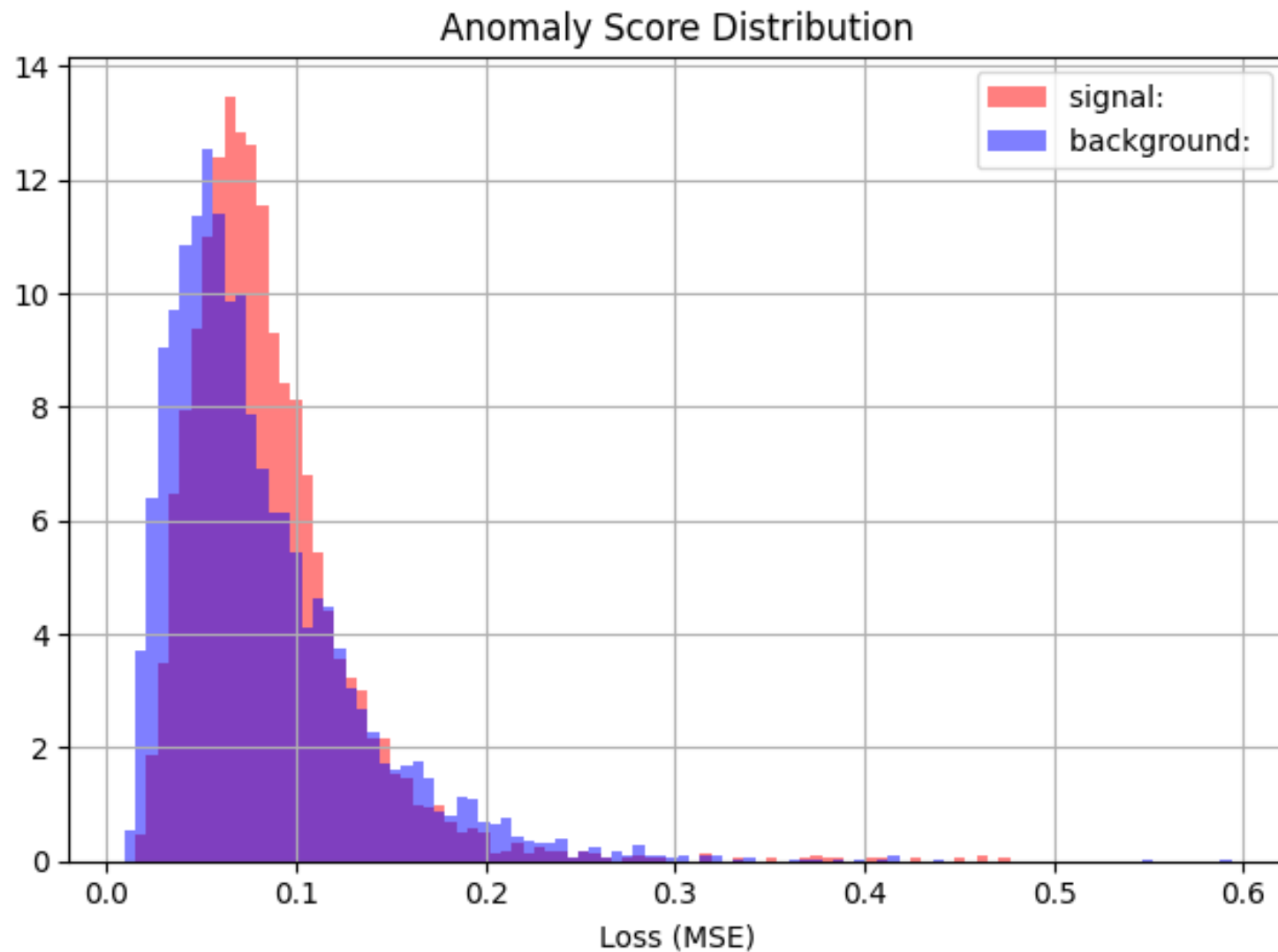
ROC

AUC = 0.500



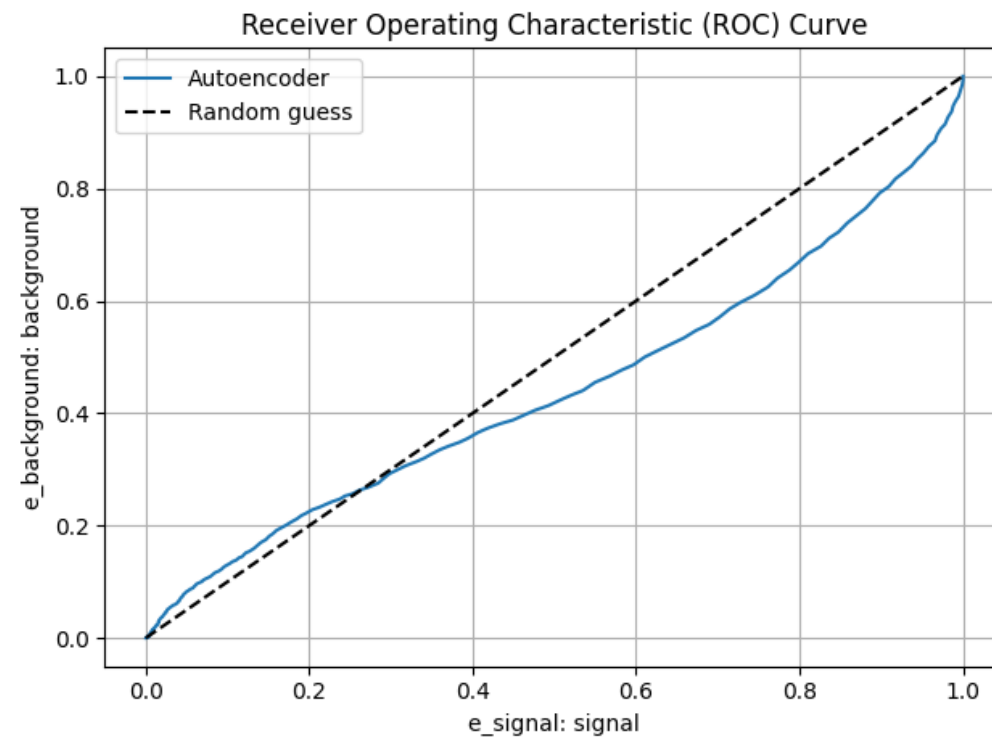
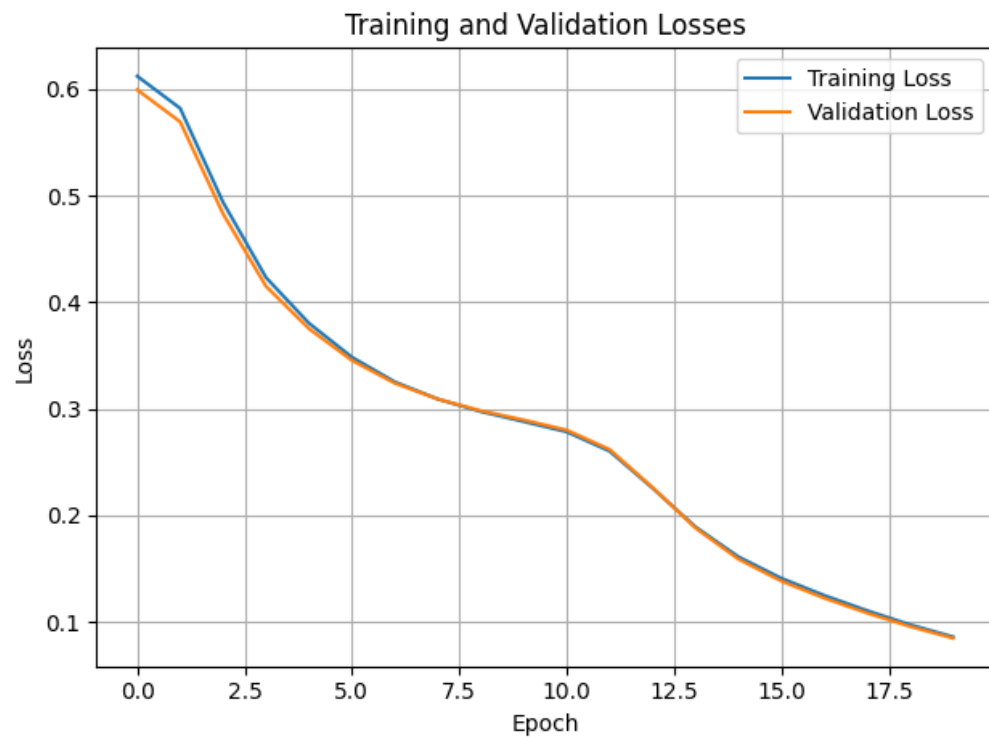
pT: 1000-
1400 GeV

Slight separation



Loss and ROC

- AUC = 0.558
- Slightly better than pT: 200-300



Next Steps:

- Combine QCD data from different files: e.g. 200-400
 - What is a good range?
- Increase number of epochs to 60-70
- Fix mass-based edges
- Test against a different signal: $W^- \rightarrow e^- \bar{\nu}_e$